

# UTD Online Programming Contest

## September 3rd 2016

### Novice Division

- Time Allowed: three hours
- **Each team must submit their attempted solutions from only one computer.** This is a PC2Team restriction.
- Answer the questions in any order.
- Use only Java, C or C++.
- Your program source code must be contained in one source file.
- Do not use the “package” construct in your Java code.
- Your programs must read from the data file given at the top of the problem description and output to System.out (cout for C/C++ programmers.)
- Make sure that your last line of output ends in a newline (i.e. is printed with System.out.println(...))
- **Do not copy solutions from the web.**
- Your solutions must be entirely yours, typed by you during the time allowed for the contest.
- As soon as you have solved a problem, submit ONLY the source file via your online PC2Team client.

## Scoring

Equal points will be awarded to each question, even though they may not be equally difficult.

In order to break ties, we will use penalty points. The penalty points for a question will be zero if the question is not answered correctly. If a correct submission occurs for a question at time T minutes, then T penalty points will be added, plus 20 penalty points for each incorrect submission for that question.

## E Got Change?

Data File: E.txt  
Time allowed = 10 seconds

Given a denomination coin set, such as  $\{1, 5, 10, 25, 50, 100\}$ , what is the smallest number of coins needed to make up 87 cents? There is a greedy algorithm for this;

Use the largest denomination as many times as possible, then repeat the process with each lower denomination until the required amount is reached.

For 87c the minimum number of coins is five:

$$0 \times 100c + 1 \times 50c + 1 \times 25c + 1 \times 10c + 0 \times 5c + 2 \times 1c$$

But if the denomination set is  $\{1, 5, 10, 12, 25, 50, 100\}$ , the greedy method will not always give the best answer. Try 16c for example.

Given an amount to be changed and a denomination set, find the minimum number of coins to make up the given amount.

### Input:

The input file begins with integer  $T < 100$ , on a line by itself, giving the number of test cases. Each test case follows on a single line and comprises space-separated integers. The amount,  $A$ ,  $A < 1000$ , to be changed comes first, followed by the denomination set, given as a series of distinct integers in increasing order of size. The largest denomination will not be larger than 100.

### Output:

For each test case print one line containing the minimum number of coins needed to make up the given amount.

| Sample Input        | Sample Output |
|---------------------|---------------|
| 3                   |               |
| 87 1 5 10 25 50 100 | 5             |
| 16 1 5 10 25 50     | 3             |
| 16 1 5 10 12 25     | 3             |

## F Tiling a Rectangle

Data File = F.txt

Time limit = 10 seconds

You are given the dimensions, *length*, *width*, of a rectangular floor. Square tiles, all of the same size, are to be used to tile the floor. Find the size of the largest tile that can be used to cover the floor completely without any tiles overlapping and without cutting any tiles. Assume that the tiles are made of marble so that no grout is needed between the tiles.

### Input:

The input file will contain up to 20 test cases. Each test case will comprise two integers,  $X, Y$   $100000 > X, Y > 0$ , space separated, on a line by itself. The input file will be terminated by End of File.

### Output:

For each test case output one line of text containing the length of the side of the largest square tile that can be used to tile the rectangle.

| Sample Input | Sample Output |
|--------------|---------------|
| 53 17        | 1             |
| 21 24        | 3             |
| 39 130       | 13            |
| 30 16        | 2             |

## G Don't Get Left With the Pepper

Data File = G.txt

Time limit = 10 seconds

Two players, Joe and Jane, play a simple game. They are presented with a large jar containing  $N$  wrapped pieces of candy and one chili pepper. They take turns, each removing one, two, or three pieces of candy from the jar. The object of the game is to leave your opponent with the chili pepper which, of course, he/she must eat.

Joe and Jane are both expert players of this game.

Given a game situation and whose turn it is to remove some candy, figure out who will win.

### Input:

The input file will begin with integer  $T$ , the number of test cases,  $T < 20$ . Each test cases comprises one line of text containing  $N$ , the number of pieces of candy left in the jar, and the name of the player that is to remove candy next.

### Output:

For each test case output one of the lines, "Joe wins" or "Jane wins".

| Sample Input | Sample Output |
|--------------|---------------|
| 2            |               |
| 53 Joe       | Joe wins      |
| 24 Jane      | Joe wins      |

## H Postal Delivery Route

Input File: H.txt  
Run Time Limit: 10 sec

Given a street map with no dead-ends, cul-de-sacs or one-way-streets, is there a route that requires the postal delivery person to traverse every street exactly once?

There are houses on only one side of each street and there is a path via the streets from any intersection to any other intersection.

Your route must begin and end at the same intersection.

### Input:

The input will begin with a line containing  $N$  ( $1 \leq N \leq 20$ ), the number of test cases. Each test case begins with a line containing two integers  $M$  ( $1 \leq M \leq 20$ ) and  $P$  ( $1 \leq P \leq 50$ ), where there are  $M$  intersections, numbered  $1, 2, \dots, M$ , and  $P$  streets.  $P$  lines follow, each one containing two integers, giving the intersection numbers at the ends of that street.

### Output:

For each test case, print the the word “yes” or “no” according to whether there is a route that meets the above requirements.

| Sample Input | Expected Output |
|--------------|-----------------|
| 3            | yes             |
| 4 4          | no              |
| 1 2          | no              |
| 2 3          |                 |
| 3 4          |                 |
| 4 1          |                 |
| 4 5          |                 |
| 1 2          |                 |
| 2 3          |                 |
| 3 4          |                 |
| 4 1          |                 |
| 1 3          |                 |
| 5 6          |                 |
| 2 1          |                 |
| 3 1          |                 |
| 4 5          |                 |
| 2 3          |                 |
| 2 4          |                 |
| 5 3          |                 |

## I Gnomes

Data File: I.txt

Time allowed = 10 seconds

In the book *All Creatures of Mythology*, gnomes are kind, bearded creatures, while goblins tend to be bossy and simple-minded. The goblins like to harass the gnomes by making them line up in groups of three, ordered by the length of their beards. The gnomes, being of different physical heights, vary their arrangements to confuse the goblins. Therefore, the goblins must actually measure the beards in centimeters to see if everyone is lined up in order.

Your task is to write a program to assist the goblins in determining whether or not the gnomes are lined up properly, either from shortest to longest beard or from longest to shortest.

### Input:

The input starts with line containing a single integer  $N$ ,  $0 < N < 30$ , which is the number of groups to process. Following this are  $N$  lines, each containing three distinct positive integers less than 100.

### Output:

There is a title line, then one line per set of beard lengths. See the sample output for capitalization and punctuation.

| Sample Input | Sample Output |
|--------------|---------------|
| 3            | Gnomes:       |
| 40 62 77     | Ordered       |
| 88 62 77     | Unordered     |
| 91 33 18     | Ordered       |

## J Alien Sort

Data File: J.txt  
Time allowed = 10 seconds

We know the normal alphabetical order of the English alphabet, and we can then sort words or other letter sequences. For instance these words are sorted:

ANTLER  
ANY  
COW  
HILL  
HOW  
HOWEVER  
WHATEVER  
ZONE

The standard rules for sorting letter sequences are used:

- The first letters are in alphabetical order.
- Among strings with the same prefix, like the prefix AN in ANTLER and ANY, they are ordered by the first character that is different, T or Y here.
- One whole string may be a prefix of another string, like HOW and HOWEVER. In this case the longer sequence comes after the shorter one.

The Gorellians, at the far end of our galaxy, have discovered various samples of English text from our electronic transmissions, but they did not find the order of our alphabet. Being a very organized and orderly species, they want to have a way of ordering words, even in the strange symbols of English. Hence they must determine their own order. Unfortunately they cannot agree, and every Gorellian year, they argue and settle on a new order.

For instance, if they agree on the alphabetical order

UVWXYZNOPQRSTHIJKLMABCDEFG

then the words above would be sorted as

WHATEVER  
ZONE  
HOW  
HOWEVER  
HILL  
ANY  
ANTLER  
COW

The first letters of the words are in their alphabetical order. Where words have the same prefix, the first differing letter determines the order, so the order goes ANY, then ANTLER,

since Y is before T in their choice of alphabet. Still HOWEVER comes after HOW, since HOW is a prefix of HOWEVER.

Dealing with the different alphabetical orders each year by hand (or tentacle) is tedious. Your job is to implement sorting with the English letters in a specified sequence.

**Input:**

The input will contain one or more datasets. Each dataset will start with a line containing an integer  $n$  and a string  $s$ , where  $s$  is a permutation of the English uppercase alphabet, used as the Gorellians' alphabet in the coming year. The next  $n$  lines ( $1 \leq n \leq 20$ ) will each contain one non-empty string of letters. The length of each string will be no more than 30. Following the last dataset is a line containing only 0.

**Output:**

The first line of output of each dataset will contain "year " followed by the number of the dataset, starting from 1. The remaining  $n$  lines are the  $n$  input strings sorted assuming the alphabet has the order in  $s$ .

| Sample Input                                                                                                       | Sample Output                                                                          |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 8 UVWXYZNOPQRSTHIJKLMABCDEFGH<br>ANTLER<br>ANY<br>COW<br>HILL<br>HOW<br>HOWEVER<br>WHATEVER<br>ZONE                | year 1<br>WHATEVER<br>ZONE<br>HOW<br>HOWEVER<br>HILL<br>ANY<br>ANTLER<br>COW           |
| 5 ZYXWVUTSRQPONMLKJIHGFEDCBA<br>GO<br>ALL<br>ACM<br>TEAMS<br>GO                                                    | year 2<br>TEAMS<br>GO<br>GO<br>ALL<br>ACM                                              |
| 10 ZOTFISENWABCDGHJKLMPQRUVXY<br>THREE<br>ONE<br>NINE<br>FIVE<br>SEVEN<br>ZERO<br>TWO<br>FOUR<br>EIGHT<br>SIX<br>0 | year 3<br>ZERO<br>ONE<br>TWO<br>THREE<br>FOUR<br>FIVE<br>SIX<br>SEVEN<br>EIGHT<br>NINE |